Scientific Python Tutorial

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This Tutorial

- This tutorial is for someone with basic python experience.
- First I begin with a few intermediate items not covered in basic tutorials.
- Then I'll provide a brief survey of the scientific packages available, including
  1. matplotlib - plotting library, like Matlab
  2. numpy - fast arrays manipulation
  3. scipy - math methods galore.
  4. ipython - better interactive python

pass

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- It's used for yet unwritten function.
- If you try to call a function that doesn't exist, you get an error.
- pass just creates a function that does nothing.
- Great for planning work!

```
def step1():
    pass
step1() # no error!
step2() # error, darn!
```

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- Use this as a default value for a variable,
- or as a return value when things don't work, and you don't want a catastrophic error.
- Test if `something is None`, to see if you need to handle these special cases

```
name=None
if name is None:
    name = 'Johnny'
```

"__main__"

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- The script/console that you are running is called `"__main__"`.
- Use this to make a file both a module *and* a script.

```
# module greeting.py
def hey_there(name):
    print "Hi!", name, "... How's it going?"

hey_there('Joey')
if __name__ == "__main__":
   hey_there('Timmy')
```

`python greeting.py` $\longrightarrow$ "Hi! Timmy ... How's it going?"

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- or the first thing inside a function, documenting it.
- This is what `help()` uses...

```
def dot_product(v1, v2):
    """Perform the dot product of v1 and v2.

    `v1` is a three element vector.
    `v2` is a three element vector.
    """
    sum = 0 #....
```

Keyword Arguments

- Arguments that have default values.
- In the function signature, `keyword=default_value`
- `None` is a good default if you want to make a decision

```
def hello(name='Joe', repeat=None):
    if repeat is None:
        repeat=len(name)
    print 'Hi!', name*repeat
```

Returning Values

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- Formally this creates a **Tuple** - an *immutable* list
- You can collect the returned values as a Tuple, or as individual values.

```
1 def pows(val):
2     return val, val*val, val*val*val
3 two, four, eight = pows(2)
4 ones = pows(1)
5 print ones[0], ones[1], ones[2] # 1 1 1
```

``string''.format()

- Strings can be paired with values to control printing.

```
>>> "Hi there {}!".format('Yann')
'Hi there Yann!'
>>> "Coords: {}, {};".format(-1, 2)
'Coords: -1, 2;'
>>> "{1}, the cost is {0:.2f}".format(1125.747025, 'Y
'Yann, the cost is 1125.75'
```

Keyword arguments to format

- It's sometimes tricky to format many values
- You can name some of the format targets

```
email = """
Subject: {subject}
Date: {mon:2d}/{day:2d}/{year:2d}
Message: {msg}
"""
print email.format(mon=10, year=12, day=31,
                   subject='Happy Halloween',
                   msg='Booh')
```

Keyword arguments to format – result

```
>>> email = """
... Subject: {subject}
... Date: {mon:2d}/{day:2d}/{year:2d}
... Message: {msg}
... """
>>> print email.format(mon=10, year=12, day=31,
...                    subject='Happy Halloween',
...                    msg='Booh')

Subject: Happy Halloween
Date: 10/31/12
Message: Booh
```

More features at:

http://docs.python.org/library/string.html

Basic File Manipulation

- You create a file object with `open('filename', 'r')`
- 'r' for reading, 'w' for writing
- `read()` for single string
- `readlines()` for a list of lines
- `close()` when you're done.

```
outfile = open('story.txt', 'w')
outfile.write('Once upon a time ...\n')
outfile.close()
```

csv reader and writer

- For reading and writing comma-separated-values
- `csv.reader` for reading, `csv.writer` for writing
- Dialects option correspond to predefined formats
- 'excel' for excel output without needing to know the separator and quote characters

```
reader = csv.reader(file)
for row in reader:
    # row is a list
writer = csv.writer(file)
writer.writerow([1,2,3])
```

argparse – easy command line arguments

- argparse module is the easiest way.
- You first create a ArgumentParser object
- You define the allowable arguments with the add_argument function
- You can add required or optional arguments
- sys.argv is a list of arguments passed to the program/script.
- Pass this list to parse_args function to process and get an object with your parameters defined.

argparse – example

Look over examples\argparse_ex.py

```
import sys
import argparse
parser = argparse.ArgumentParser(description='Plot zombie statisti
parser.add_argument('prefix', help='the prefix for the plot filena
parser.add_argument('-p', dest='pop', default=500, type=int,
                    help='Set the startring population')
parser.add_argument('-s', dest='show',
                    help='Show the figure', action='store_true')
parser.add_argument('city', help='Plot information for a specific
                    nargs='?', default=None)

args = sys.argv[1:]
params = parser.parse_args(args)

print "prefix = ",params.prefix
print "pop = ",params.pop
if params.city is not None:
    print "city = ", params.city
print "show? "
if params.show:
```

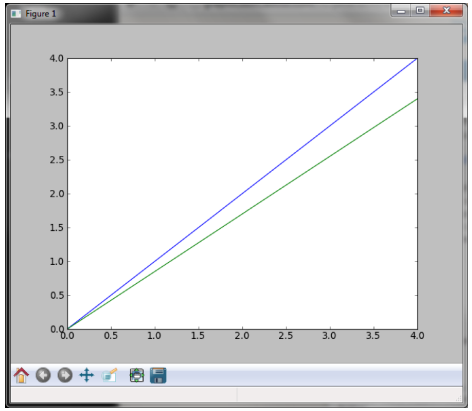
subprocess – running external programs

```
1 import subprocess
2 output = subprocess.call(['ls', '-1'])
3 print "output = ",output
```

- subprocess provides many tools
- The most basic is the call function.
- It takes a list that is joined and executed.
- output just holds the exit code (0 if successful)
- check_output is like call but 1) returns output and 2) causes an error if program fails

Plotting in scripts

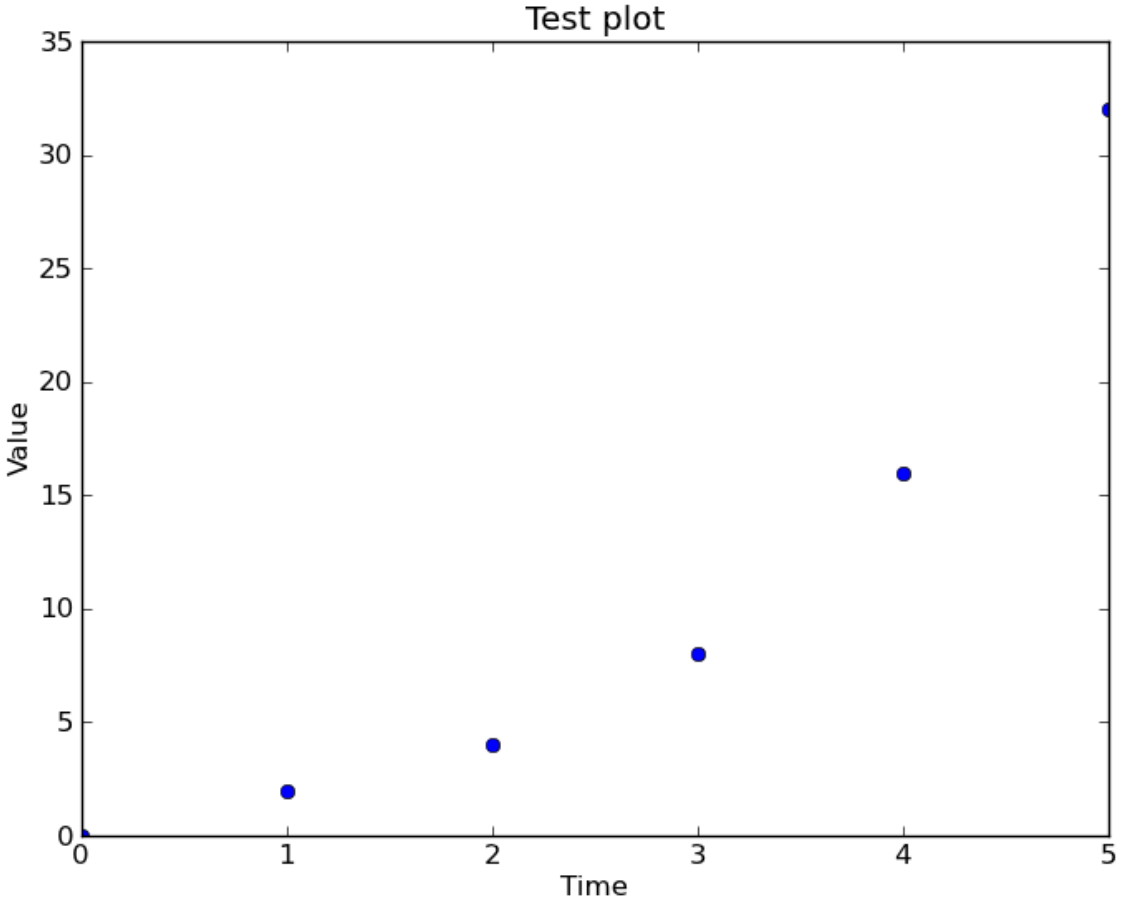
- `matplotlib` is a package that has many modules, `pyplot` is the main driver.
- `matplotlib` is designed for both interactive and script-based use.
- a `Figure` can contain many `Axes`s which contain many plots.
- `pyplot` allows you to access a default `Figure` and `Axes`.



Basic Example

```
1 from matplotlib import pyplot
2
3 pyplot.plot([0, 2, 4, 8, 16, 32], "o")
4
5 pyplot.ylabel("Value")
6 pyplot.xlabel("Time")
7 pyplot.title("Test plot")
8
9 pyplot.show()
```

Basic Plot - Results



World Population - Practice 1

practice/world_population.py

```
def get_data():
    data = file("world_population.txt", "r").readlines()
    dates = []
    populations = []
    for point in data:
        date, population = point.split()
        dates.append(date)
        populations.append(population)
    return dates, populations
```

1. We read in the data

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        populations.append(population)
    return dates, populations
```

1. We read in the data
2. For each line ('point') we need to separate the date from population
3. `split()` splits text at any whitespace, by default.

World Population - Practice 2

```
def plot_world_pop(dates, populations):
    pass

dates, populations = get_data()
plot_world_pop(dates, populations)
pyplot.show()
```

Finish this method:

1. Make a plot of population (y) vs dates (x)
2. Title: "World population over time"
3. Y-axis: "World population in millions"
4. X-axis: "Year"
5. Set the plot type to a Magenta, downward-triangle

World Population - Getting Help

http://matplotlib.sourceforge.net/ → "Quick Search" for "pyplot.plot" → "matplotlib.pyplot.plot"

`http://matplotlib.sourceforge.net/api/pyplot_api.html#matplotlib.pyplot.plot`

matplotlib.sourceforge.net/api/pyplot_api.html#matplotlib.pyplot.plot

matplotlib.pyplot.**plot**(**args, **kwargs*)

Plot lines and/or markers to the Axes. *args* is a variable length argument, allowing for multiple *x*, *y* pairs with an optional format string. For example, each of the following is legal:

```
plot(x, y)        # plot x and y using default line style and color
plot(x, y, 'bo')  # plot x and y using blue circle markers
plot(y)           # plot y using x as index array 0..N-1
plot(y, 'r+')     # ditto, but with red plusses
```

If *x* and/or *y* is 2-dimensional, then the corresponding columns will be plotted.

An arbitrary number of *x*, *y*, *fmt* groups can be specified, as in:

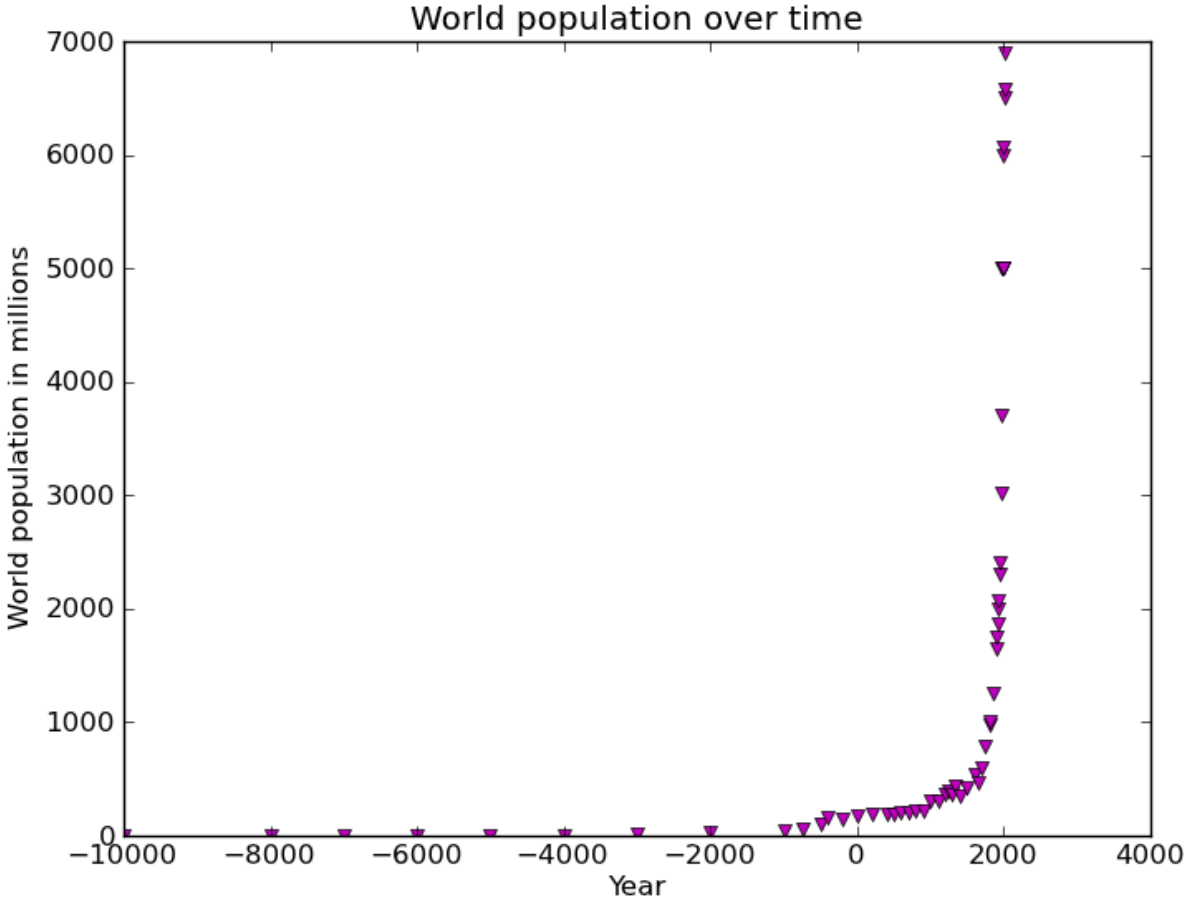
```
a.plot(x1, y1, 'g^', x2, y2, 'g-')
```

Return value is a list of lines that were added.

The following format string characters are accepted to control the line style or marker:

character	description
'-'	solid line style
'--'	dashed line style
'-.'	dash-dot line style
':'	dotted line style
'.'	point marker
','	pixel marker
'o'	circle marker
'v'	triangle_down marker
'^'	triangle_up marker
'<'	triangle_left marker
'>'	triangle_right marker
'1'	tri_down marker
'2'	tri_up marker
'3'	tri_left marker
'4'	tri_right marker
's'	square marker

World Population - Final



Life Expectancy - Practice 1

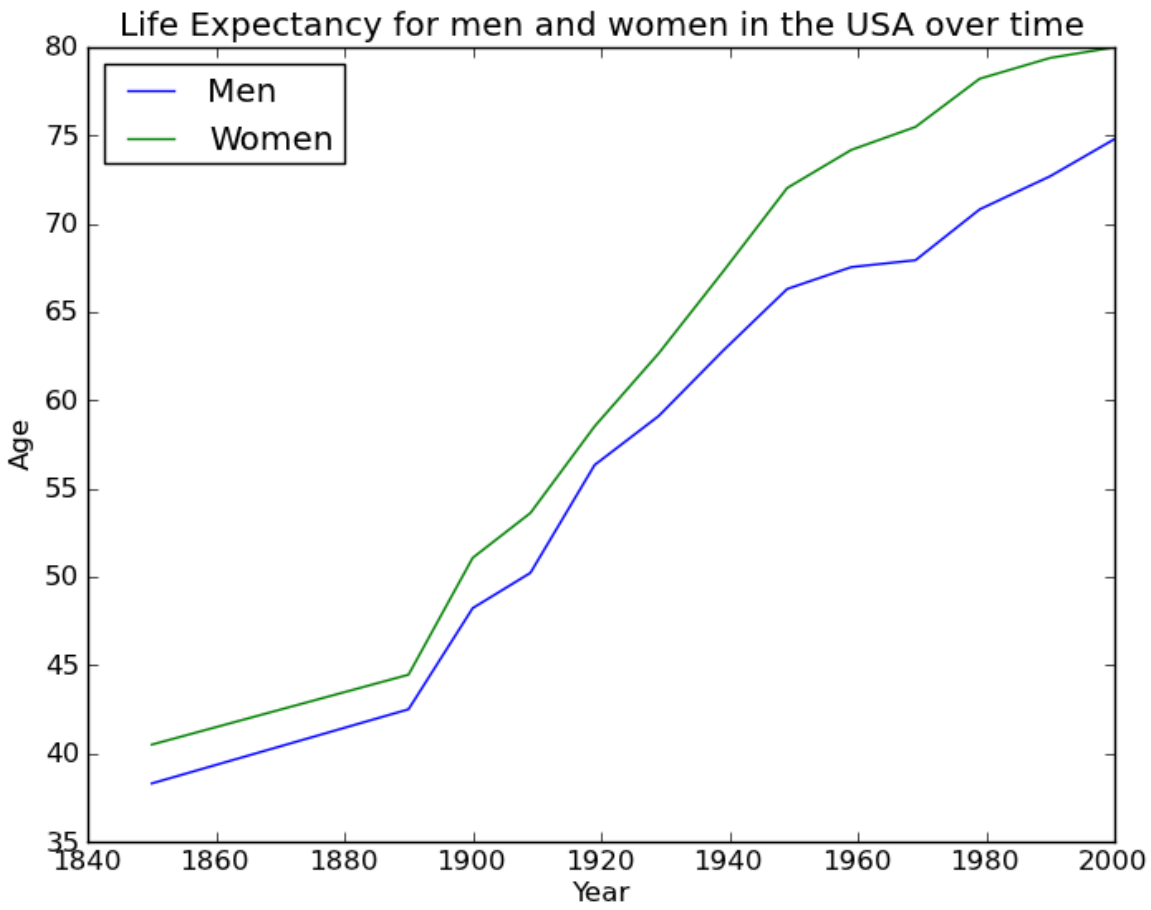
practice/life_expectacies_usa.py

```
1  from matplotlib import pyplot
2
3  def get_data():
4      # each line: year,men,women
5      data = file("life_expectancies_usa.txt", "r").readlines()
6      dates = []
7      men = []
8      women = []
9      # finish me!
10     return dates, men, women
11
12 def plot_expectancies(dates, men, women):
13     pass
14
15 dates, men, women = get_data()
16 plot_expectancies(dates, men, women)
17 pyplot.show()
```

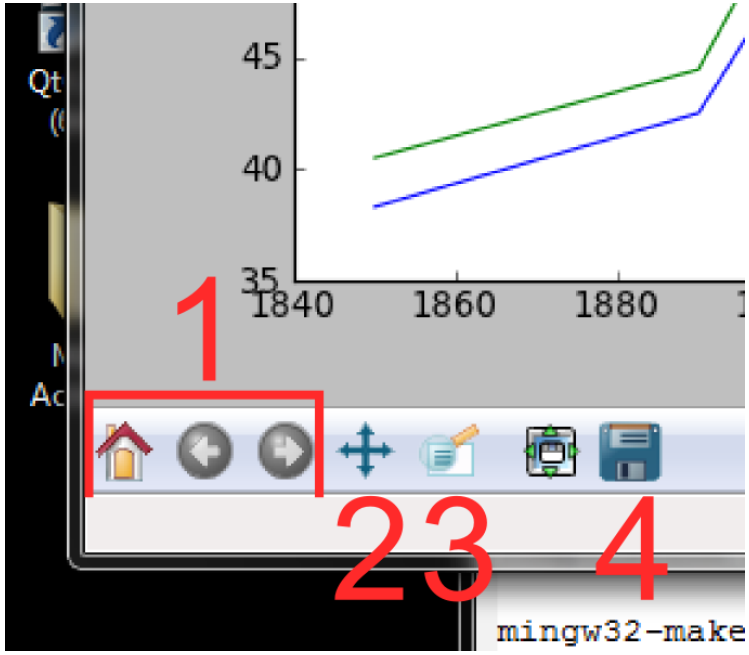
Life Expectancy - Practice 2

1. Use `split(',')` to split strings at commas
2. Add a label to each plot (look at documentation)
3. Label Axes and give a title.
4. Call plot 2x to plot two lines
5. Add a legend: `pyplot.legend`
   `http://matplotlib.sourceforge.net` → search for "pyplot.legend"

Life Expectancy - Results



Plotting Interactively



1. **Home** - **Backward** - **Foward** - Control edit history

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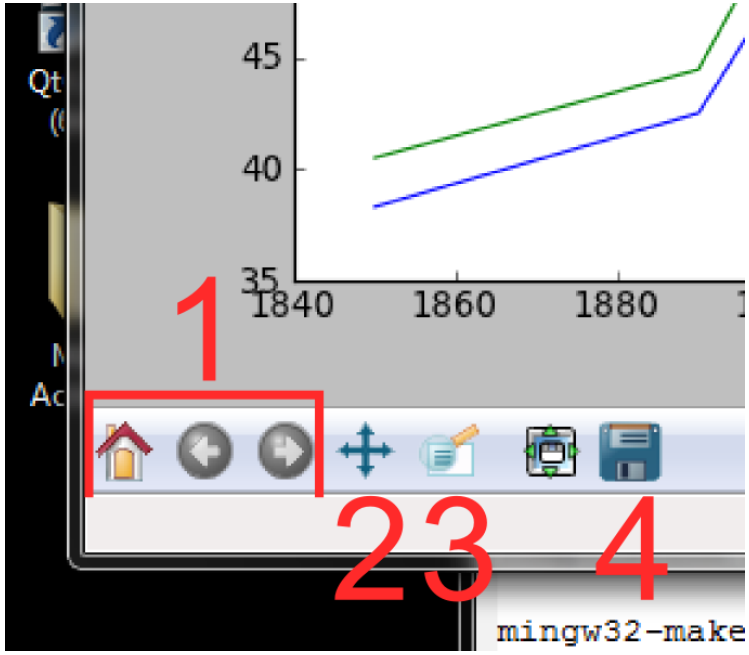

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2. **Pan** - **Zoom** - Left click + drag shifts center, right click + drag changes zoom

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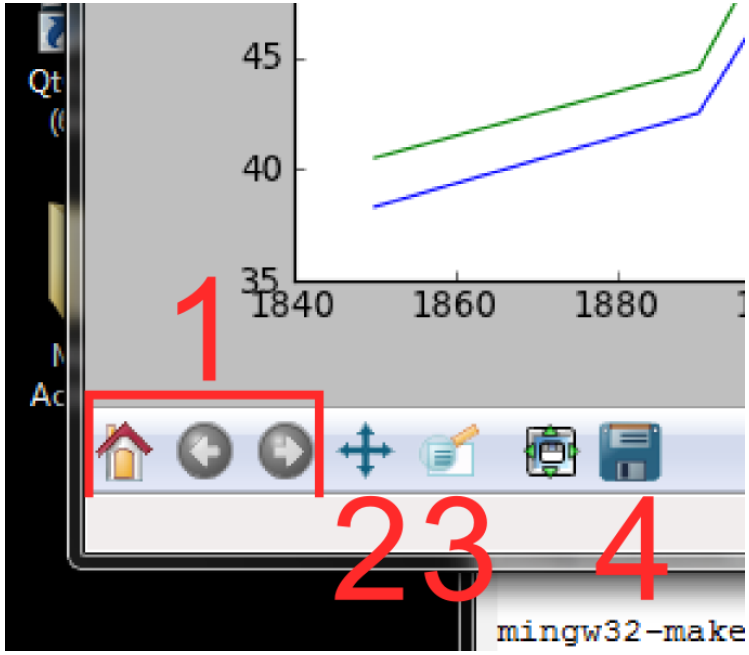

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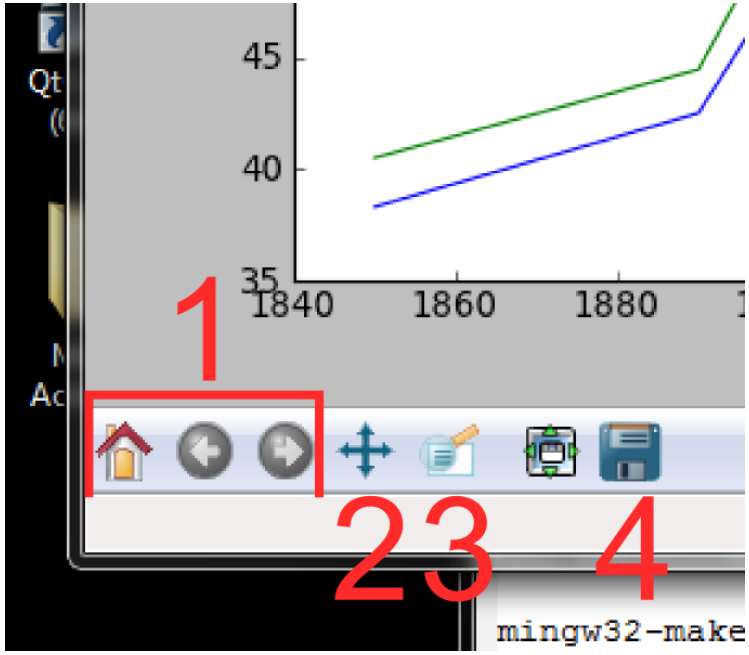

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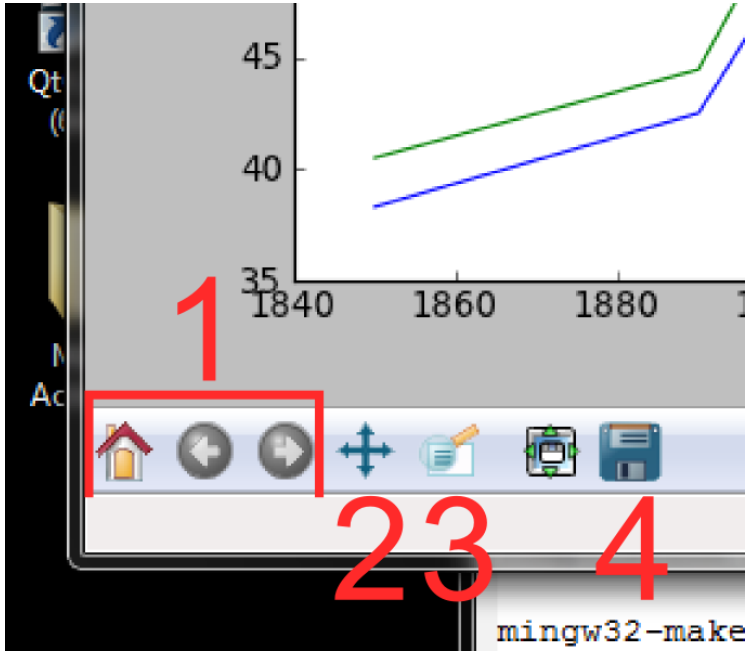

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4. **Save**

`pyplot.savefig('filename')` is an alternative to `pyplot.show()` when you are using pyplot non-interactively.

Matplotlib Resources

- **Possibilities:**
  matplotlib.sourceforge.net/gallery.html
  http://matplotlib.org/basemap/users/examples.html
- **Guide/Tutorial:**
  matplotlib.sourceforge.net/users/index.html
- **Questions:**
  stackoverflow.com
  Or contact us: scv@bu.edu
- **Source for tutorial:**
  openhatch.org/wiki/Matplotlib

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- `np.ones(5)` makes `array([ 1., 1., 1., 1., 1.])`
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- `np.ones(5)` makes `array([ 1., 1., 1., 1., 1.])`
- `np.zeros` or `np.arange`, what do they do?
- `np.linspace` is similar to `np.arange` but pass number of elements, *not* step size

Array Basics

```
import numpy as np
x = np.array(((1, 2, 3), (4, 5, 6)))
x.size # total number of elements
x.ndim # number of dimensions
x.shape # number of elements in each dimension
x[1,2] # first index is the rows, then the column
x[1] # give me '1' row
x[1][2]
x.dtype
```

Array Basics - Results

```
>>> import numpy as np
>>> x = np.array(((1, 2, 3), (4, 5, 6)))
>>> x.size # total number of elements
6
>>> x.ndim # number of dimensions
2
>>> x.shape # number of elements in each dimension
(2L, 3L)
>>> x[1,2] # first index is the rows, then the column
6
>>> x[1] # give me '1' row
array([4, 5, 6])
>>> x[1][2]
6
>>> x.dtype
dtype('int32')
```

Basic Operations

```
x = np.array([0, np.pi/4, np.pi/2])
np.sin(x)
np.dot([2, 2, 2], [2, 2, 2])
```

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```

```
>>> x = np.array([0, np.pi/4, np.pi/2])
>>> np.sin(x)
array([ 0.        ,  0.70710678,  1.        ])
>>> np.dot([2, 2, 2], [2, 2, 2])
12
```

Solve Laplace's Equation 1a

- Solve $\nabla^2 u = 0$

Solve Laplace's Equation 1a

- Solve $\nabla^2 u = 0$
- Solve iteratively, each time changing u with following equation:
  $u_{j,l}^{n+1} = 1/4(u_{j+1,l}^{n} + u_{j-1,l}^{n+1} + u_{j,l+1}^{n} + u_{j,l-1}^{n+1})$
- Just an averages of the neighboring points.

Solve Laplace's Equation 1a

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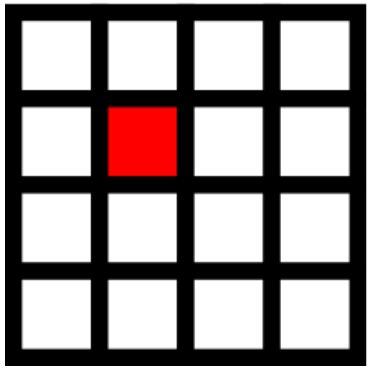
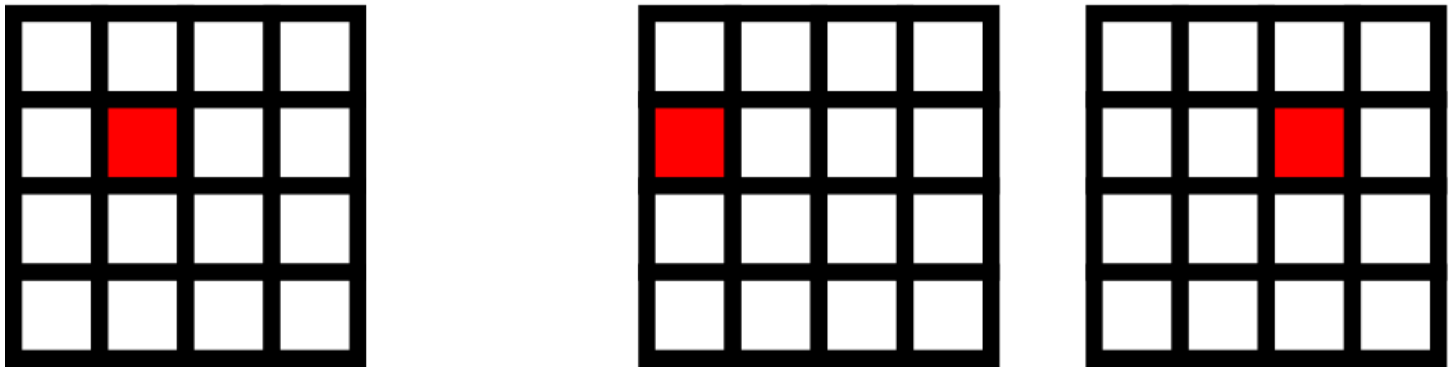
$$U_{j,l}^{n+1} = 1/4\ ($$



Solve Laplace's Equation 1b

- Solve $\nabla^2 u = 0$
- Solve iteratively, each time changing u with following equation:
  $u_{j,l}^{n+1} = 1/4(u_{j+1,l}^{n} + u_{j-1,l}^{n+1} + u_{j,l+1}^{n} + u_{j,l-1}^{n+1})$
- Just an averages of the neighboring points.

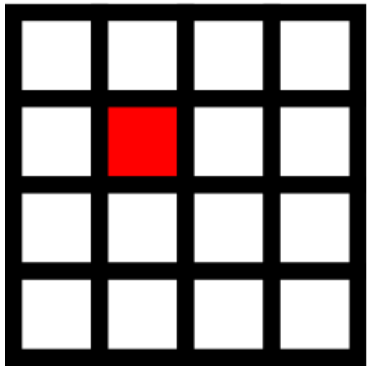
$$U_{j,l}^{n+1} = 1/4 \left(\quad U_{j-1,l}^{n} \quad + \quad U_{j+1,l}^{n} \quad + \ldots \right.$$

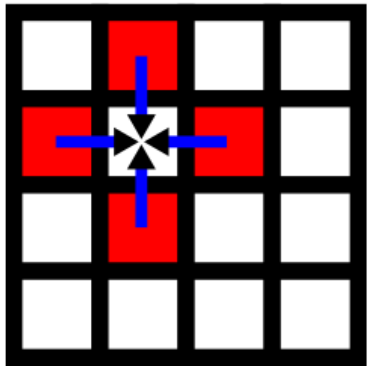


Solve Laplace's Equation 1c

- Solve $\nabla^2 u = 0$
- Solve iteratively, each time changing u with following equation:
  $u_{j,l}^{n+1} = 1/4(u_{j+1,l}^{n} + u_{j-1,l}^{n+1} + u_{j,l+1}^{n} + u_{j,l-1}^{n+1})$
- Just an averages of the neighboring points.

$$U_{j,l}^{n+1} = 1/4 \; (\quad \ldots \quad)$$





Solve Laplace's Equation 1d

- Solve $\nabla^2 u = 0$
- Solve iteratively, each time changing u with following equation:
  $u_{j,l}^{n+1} = 1/4(u_{j+1,l}^{n} + u_{j-1,l}^{n+1} + u_{j,l+1}^{n} + u_{j,l-1}^{n+1})$
- Just an average of the neighboring points.
- Repeat this calculation until $rms(u^{n+1} - u^n) < \epsilon$, some threshold
- Set some limit on total number of iterations
- `practice/laplace.py`

Solve Laplace's Equation 2

First task:

```
def pure_python_step(u):
    '''Pure python implementation of Gauss-Siedel method. Performs
    iteration.'''
    rms_err = 0
    # use for loop to loop through each element in u
    #     temporarily store old value to later calculate rms_err
    #     update current u using 4 point averaging
    #     update running value of rms_err
    # when done looping complete rms_err calculation and return it
    return rms_err
```

Slicing - 1

```
x = np.array((1, 2, 3, 4, 5, 6))
x[2]
x[2:5]
x[2:-1]
x[:5]
x[:5:2] = 10
x
```

Slicing - 2

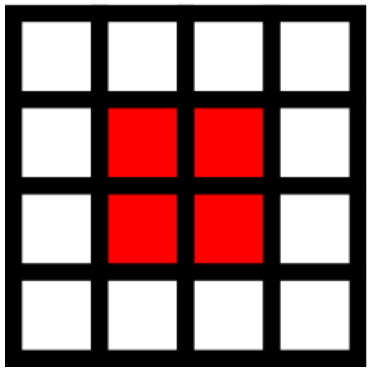
```
>>> x = np.array((1, 2, 3, 4, 5, 6))
>>> x[2]
3
>>> x[2:5]
array([3, 4, 5])
>>> x[2:-1]
array([3, 4, 5])
>>> x[:5]
array([1, 2, 3, 4, 5])
>>> x[:5:2] = 10
>>> x
array([10,  2, 10,  4, 10,  6])
```

Array Copying

```
>>> a = np.array((1,2,3,4))
>>> b = a
>>> b is a
True
>>> c = a.view()
>>> c.shape = 2,2
>>> c[1,1]=10
>>> c
array([[ 1,  2],
       [ 3, 10]])
>>> a
array([ 1,  2,  3, 10])
>>> d = a.copy()
```

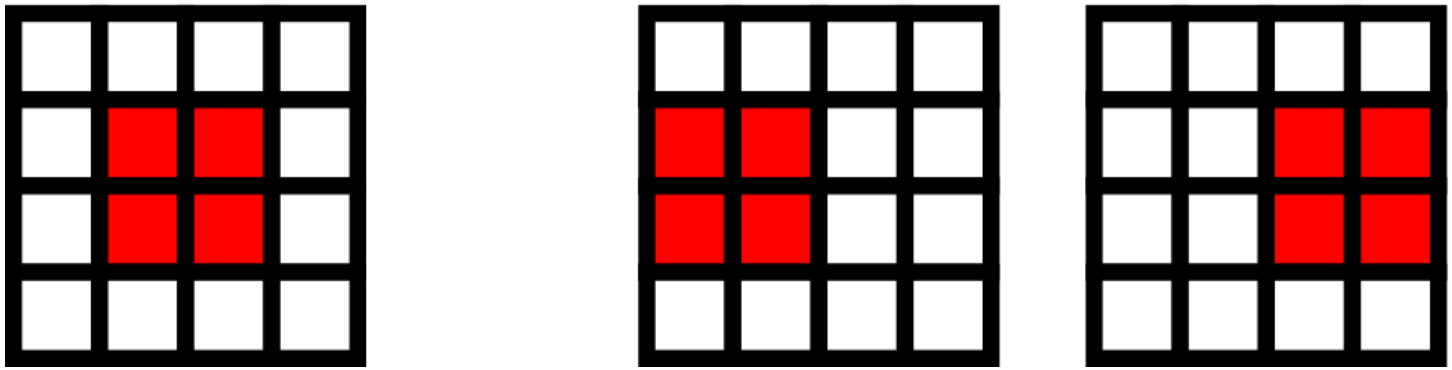
Solve Laplace's Equation 3a

$$U_{j,l}^{n+1} = 1/4 \; ($$



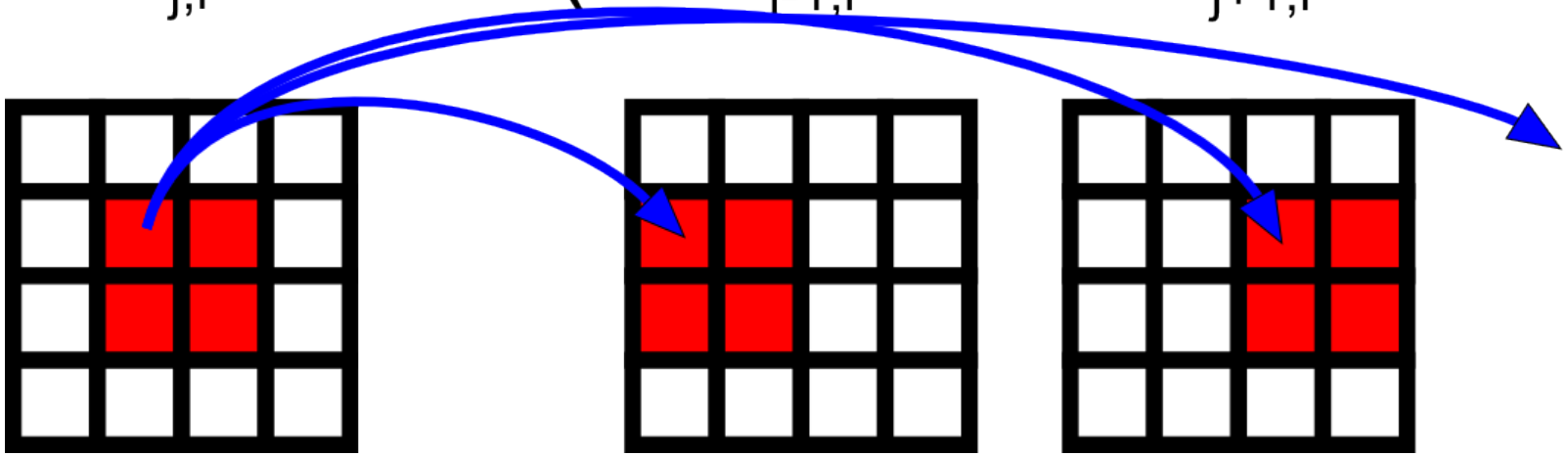
Solve Laplace's Equation 3b

$$U^{n+1}_{j,l} = 1/4 \left(U^{n}_{j-1,l} + U^{n}_{j+1,l} + \ldots \right.$$



Solve Laplace's Equation 3c

$$U^{n+1}_{j,l} = 1/4 \left(U^{n}_{j-1,l} + U^{n}_{j+1,l} + \ldots \right.$$



Solve Laplace's Equation 3d

Second task:

```
def numpy_step(u):
    '''Numpy based Jacobi's method. Performs one iteration.'''
    # make a copy so that you can calculate the error
    u_old = u.copy()
    # use slicing to shift array
    # utmp = u[1:-1, 1:-1] makes a new array, so that utmp[0,0] is
    # as u[1,1]
    # then
    # utmp = u[0:-2, 1:-1] makes a new array that leads to a shift
    # because utmp[0,0] is the same as u[0, 1]
    # use this concept to solve this equation in on line
    # u = 1/4*(u_{j-1,i} + u_{j+1,i} + u_{j, i-1} + u_{j, i+1})
    return calc_err(u, u_old)
```

Solve Laplace's Equation 4a

Third task:

```
def laplace_driver(u, stepper, maxit=100, err=1e3):
    '''Repeatedly call stepper(u) to solve nabla^2 u = 0 until
    rms error < err or maxit number of iterations is reached.

    `u` - a numpy array
    `stepper` - a function whose sole argument is u
    '''
    rms_err = 0
    # take one step with stepper, to define initial rms_err
    # loop until rms_err < err
    #     check to see that number of iterations is less than maxi
    #     perform single iteration using stepper method
    # return rms_error
    return rms_err
```

Solve Laplace's Equation 4b

```
59 def time_method(stepper):
60     '''Time how long a particular stepper takes to solve an ideal
61     u = set_bc(np.zeros((100,100)))
62     start = time.time()
63     err = laplace_driver(u, stepper)
64     return time.time()-start
65
66 if __name__ == "__main__":
67     pure_python_time = time_method(pure_python_step)
68     numpy_time = time_method(numpy_step)
69     print "Pure python method takes {:.3f} seconds".format(pure_py
70     print "Numpy method takes {.3f} seconds".format(numpy_time)
```

Solve Laplace's Results

```
Pure python method takes 3.624 seconds
Numpy method takes 0.016 seconds
```

Plenty More

- www.scipy.org/Tentative_NumPy_Tutorial
- "Universal Functions" section
- PyTrieste Numpy Tutorial

scipy: Many Useful Scientific Tools

http://docs.scipy.org/doc/scipy/reference/tutorial/index.html

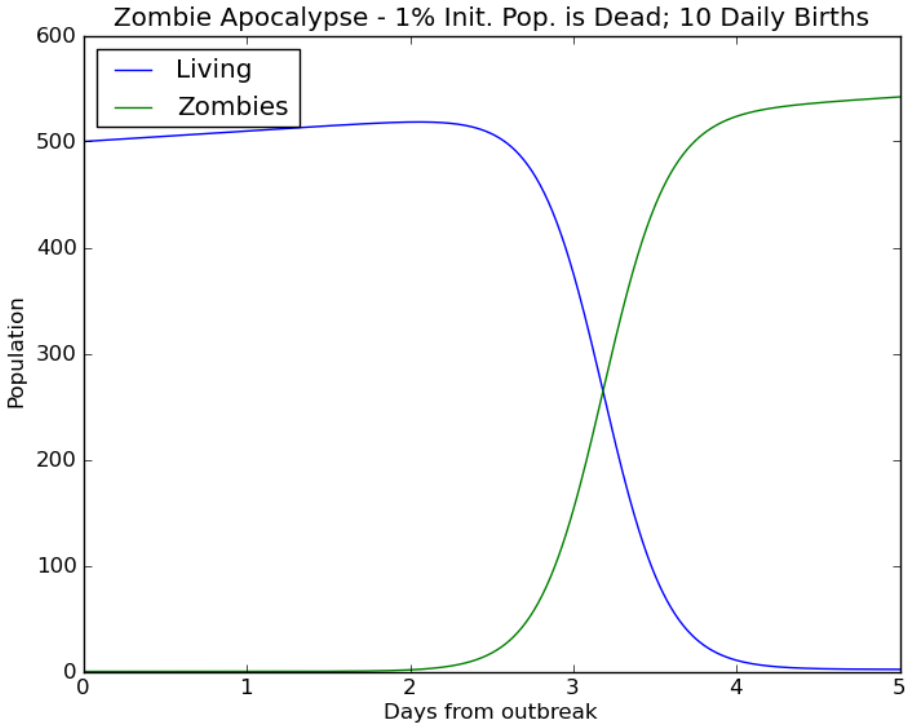
- Numpy
- Integration
- Optimization
- Interpolation
- FFT
- Linear Algebra
- Statistics
- And much more....

Constants

```
>>> from scipy import constants
>>> constants.c # speed of light
299792458.0
>>> # physical constants: value, units, uncertainty
>>> constants.physical_constants["electron mass"]
(9.10938291e-31, 'kg', 4e-38)
>>> # look-up constants, only first 3 !
>>> masses = constants.find("mass")[:3]
>>> for val in masses:
...     print "'{}' is available".format(val)
...
'Planck mass' is available
'Planck mass energy equivalent in GeV' is available
'alpha particle mass' is available
```

Zombie Apocalypse - ODEINT

http://www.scipy.org/Cookbook/Zombie_Apocalypse_ODEINT



Zombie Apocalypse - ODEINT

enerated/scipy.integrate.odeint.html

on » SciPy v0.11.dev Reference Guide (DRAFT) » Integration and ODEs (scipy.integrate) » previous | next | modules | modules | index

scipy.integrate.odeint

`scipy.integrate.odeint`(*func, y0, t, args=(), Dfun=None, col_deriv=0, full_output=0, ml=None, mu=None, rtol=None, atol=None, tcrit=None, h0=0.0, hmax=0.0, hmin=0.0, ixpr=0, mxstep=0, mxhnil=0, mxordn=12, mxords=5, printmessg=0*) [source]

Integrate a system of ordinary differential equations.

Solve a system of ordinary differential equations using lsoda from the FORTRAN library odepack.

Solves the initial value problem for stiff or non-stiff systems of first order ode-s:

```
dy/dt = func(y,t0,...)
```

where y can be a vector.

Parameters :

func : callable(y, t0, ...)
Computes the derivative of y at t0.

y0 : array
Initial condition on y (can be a vector).

t : array
A sequence of time points for which to solve for y. The initial value point should be the first element of this sequence.

args : tuple

Zombie Apocalypse - ODEINT

Look at `examples\zombie.py`

```
def calc_rate(P=0, d=0.0001, B=0.0095, G=0.0001, A=0.
    def f(y, t):
        Si = y[0]
        Zi = y[1]
        Ri = y[2]
        # the model equations (see Munz et al. 2009)
        f0 = P - B*Si*Zi - d*Si
        f1 = B*Si*Zi + G*Ri - A*Si*Zi
        f2 = d*Si + A*Si*Zi - G*Ri
        return [f0, f1, f2]
    Z0 = 0                          # initial zombie populati
    R0 = 0                          # initial death populatio
    y0 = [S0, Z0, R0]               # initial condition vecto
    t  = np.linspace(0, 5., 1000)   # time grid
    # solve the DEs
    soln = odeint(f, y0, t)
```

ipython (1) - plotting

Amazing interactive python.
Using the '-pylab' flag, you can get plotting for free

```
% ipython -pylab
...
In [1]: plot(range(10))
```

ipython (2) - Magic Functions

Amazing interactive python.
It provids "Magic" functions:

- `cd` - like unix change directory
- `ls` - list directory
- `timeit` - time execution of statement
- and so on ...

ipython (3) - Logging

Amazing interactive python.
You can log an interactive session:

```
In [1]: logstart mylogfile.py
```

Everything you type will be recorded to mylogfile.py
To re-run it in ipython, us the **run** command:

```
In [1]: run -i mylogfile.py
```

ipython (4) - Misc.

Amazing interactive python.

- You can use ? instead of `help()`.
  `In [1]: len?` and Enter will print help
  In fact you get even more information than the standard `help()`
- Tab-completion
- Start typing a mtehod, and "pop-up" help is shown
  You need a newer version than Katana's, and you need to type `ipython qtconsole`

Resources

- SCV help site
- Numpy Tutorial
- Scipy tutorial
- Scientific Python Tutorial
- Another Scientific Python Tutorial
- iPython Tutorial

What I Didn't Cover

- Exception Handling
- Duck Typing
- Virtualenv
- Pandas
- mpi4py
- And many other useful things